

```
!pip install gensim
import gensim.downloader as api
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.manifold import TSNE
```

Collecting gensim

Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl.metadata (8 Requirement already satisfied: numpy>=1.18.5 in /usr/local/lib/python3.12/dist-packages (from gensim) Requirement already satisfied: scipy>=1.7.0 in /usr/local/lib/python3.12/dist-packages (from gensim) Requirement already satisfied: smart_open>=1.8.1 in /usr/local/lib/python3.12/dist-packages (from gensim) Requirement already satisfied: wrapt in /usr/local/lib/python3.12/dist-packages (from smart_open>=1.8.1) Downloading gensim-4.4.0-cp312-cp312-manylinux_2_24_x86_64.manylinux_2_28_x86_64.whl (27.9 MB)
27.9/27.9 MB 53.1 MB/s eta 0:00:00

Installing collected packages: gensim
Successfully installed gensim-4.4.0

```
model = api.load("glove-wiki-gigaword-100") # 100-dim GloVe
print("Vocabulary size:", len(model.key_to_index))
print("Vector for 'computer':", model['computer'])
```

```
[=====] 100.0% 128.1/128.1MB downloaded
Vocabulary size: 400000
Vector for 'computer': [-1.6298e-01  3.0141e-01  5.7978e-01  6.6548e-02  4.5835e-01 -1.5329e-01
 4.3258e-01 -8.9215e-01  5.7747e-01  3.6375e-01  5.6524e-01 -5.6281e-01
 3.5659e-01 -3.6096e-01 -9.9662e-02  5.2753e-01  3.8839e-01  9.6185e-01
 1.8841e-01  3.0741e-01 -8.7842e-01 -3.2442e-01  1.1202e+00  7.5126e-02
 4.2661e-01 -6.0651e-01 -1.3893e-01  4.7862e-02 -4.5158e-01  9.3723e-02
 1.7463e-01  1.0962e+00 -1.0044e+00  6.3889e-02  3.8002e-01  2.1109e-01
 -6.6247e-01 -4.0736e-01  8.9442e-01 -6.0974e-01 -1.8577e-01 -1.9913e-01
 -6.9226e-01 -3.1806e-01 -7.8565e-01  2.3831e-01  1.2992e-01  8.7721e-02
 4.3205e-01 -2.2662e-01  3.1549e-01 -3.1748e-01 -2.4632e-03  1.6615e-01
 4.2358e-01 -1.8087e+00 -3.6699e-01  2.3949e-01  2.5458e+00  3.6111e-01
 3.9486e-02  4.8607e-01 -3.6974e-01  5.7282e-02 -4.9317e-01  2.2765e-01
 7.9966e-01  2.1428e-01  6.9811e-01  1.1262e+00 -1.3526e-01  7.1972e-01
 -9.9605e-04 -2.6842e-01 -8.3038e-01  2.1780e-01  3.4355e-01  3.7731e-01
 -4.0251e-01  3.3124e-01  1.2576e+00 -2.7196e-01 -8.6093e-01  9.0053e-02
 -2.4876e+00  4.5200e-01  6.6945e-01 -5.4648e-01 -1.0324e-01 -1.6979e-01
 5.9437e-01  1.1280e+00  7.5755e-01 -5.9160e-02  1.5152e-01 -2.8388e-01
 4.9452e-01 -9.1703e-01  9.1289e-01 -3.0927e-01]
```

```
words = ["dog", "cat", "lion", "tiger",
         "apple", "banana", "orange", "grape",
         "computer", "laptop", "keyboard", "internet",
         "paris", "london", "berlin", "tokyo"]
```

```
vectors = np.array([model[w] for w in words])
```

```
tsne = TSNE(n_components=2, random_state=42, perplexity=15)
word_coords = tsne.fit_transform(vectors)
```

```
plt.figure(figsize=(10,8))
for i, word in enumerate(words):
    x, y = word_coords[i]
    plt.scatter(x, y)
    plt.text(x+0.02, y+0.02, word, fontsize=9)
plt.title("t-SNE Visualization of Word Embeddings")
plt.show()
```

